

AKADEMIKA

BEL-ADT

ANALOG AND DIGITAL TRAINER

EXPERIMENTAL MANUAL

.... A MESSAGE FROM



Today's system designers are faced with tomorrow's problems. Basic electronics is one of the important subject need to teach while learning electronics. It is our vision to provide you with the product you need for training ensuring lasting reliability & quality.

OUR MOTTO;

- Light years ahead – refers to leadership.

As leaders in our industry in India, We are totally committed to servicing as the standard against which all are measured in the areas of

• Design • Quality • Value • Delivery • Support.

We are truly light years ahead of our competition in this area. That means that you our valued customers are guaranteed satisfaction.

As you will move through this manual you will quickly discover that we have a complete, truly innovative & superior training products we are so committed to quality that we back our products with a complete comprehensive warranty.

AKADEMIKA LAB SOLUTIONS

Sr. No. 15/8/1, Unit No.9, Kruti Industrial Estate, Opp.

Karishma Society, Off Karve Road, Kothrud, Pune,

Maharashtra- 411038

INDIA

Contact No.:+91-7447438443

Email: lab@akademika.in

www.akademika.in

SAFETY RULES

- Read carefully and follow the instructions mentioned in this manual. This user manual includes all the important points about the installation, use and the maintenance of the product. Keep this manual always with you, for quick reference.
- After unpacking the product, arrange all the accessories in proper order, so that their integrity is checked with the packing list. Also, ensure that the accessories have no visible damage.
- Before connecting the power supply to the kit, be sure that the jumpers and the connecting chords are connected correctly, as per the experiment.
- This kit must be employed only for the use for which it has been conceived, i.e. as educational kit and must be used under the direct survey of expert personnel. Any other use is inadvisable and dangerous too. The manufacturer cannot be considered responsible for eventual damages due to improper, wrong or unreasonable uses.
- In case of any fault or malfunctioning in the trainer kit, turn off the power supply. Please do not tamper the kit. If you require our service, kindly contact the service centre for technical assistance.
- BEL-ADT is High Level, High Quality Analog-Digital Trainer kit. It combines all essential functions of Analog and Digital Experiment with removable breadboard, Includes DC power supply, Function Generator, Two Pulse Switches, 8 ohm 0.25W Speaker
- The kits are liable to malfunction/under-perform if they are not operated under standard environmental conditions of temperature and humidity.

WARRANTY

This kit is warranted against defects in workmanship and materials. Any failure due to defect in either workmanship or material should occur under normal use within a year from the original date of purchase, such failure will be corrected free of charge to the purchaser by repair or replacement of defective part or parts. When the failure is result of user's neglect, natural disaster or accident, we would charge for repairs, regardless of the warranty period. The warranty does not cover include perishable items like connecting chords, crystals, etc. and other imported items.

Conditions and Limitations

The warranty is void and inapplicable if the defective product is not brought or sent to our authorized service center or sales outlet within the warranty period. Defective product will be Akademia Lab Solutions sole judgment. The defective product will be replaced with a new one or repaired, without charge or with charge.

In the warranty period if the service is needed, the purchaser should get in touch with the service center or the sales outlet. The purchaser should return the product to the service center or the sales outlet at his or her sole expense. The loss and damage in transit will be outside the preview of this warranty. A returned product must be accompanied by a written description of the defects. Type and Model No. of the kit has to be mentioned specifically. We return the product to the purchaser at our expense. In case the warranty does not cover the product on Akademia Lab Solutions judgment, we would repair the product after obtaining prior permission from purchaser who would receive an estimate statement about the repairing charges. In such cases, Akademia Lab Solutions bares the transporting expenses required to send back all the repaired products for the moment, and then repairs and transporting expenses will be charged against the purchaser by the statement of accounts.

When the authorized sales agents sell our products, they must notify the purchaser of the warranty contents, but they have no rights to stretch the meaning of original warranty contents or to offer an additional warranty. Akademia Lab Solutions does not provide any other promise or suggestive warranty and hold no liability for the damage caused by negligence, abnormal use or natural disaster. Akademia Lab Solutions is not responsible for the damages even if it is notified of above dangers in advance as well.

For more special service or overall repairs, maintenance and up gradation of products, be sure to contact our service center or the sales outlet.

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TECHNICAL SPECIFICATIONS

- **Power Supply**

Dual DC 0 To 30v, 0 To 30v Variable Positive On-Board Power Supply.

Fixed DC +5v and +12v power supply.

Single Negative Variable 0-30v Fixed Power Supply.

-5v-0-5v, +12v-0-12v AC Power Supply.

- **POT Bank**

Variable Potentiometer -2 (1k and 10k)

Function Generator: Wave from Sine, Triangular, Square and TTL

Sine Wave: - Frequency from 1Hz to 1MHz in 5 Steps and Amplitude from 0 to 6Vp-P.

Triangular Wave :- Frequency From 1hz To 800 KHZ In 5 Steps And Amplitude From 0 To 6Vp-P.

Square Wave: - Frequency from 1 Hz to 800 KHZ in 5 Steps and Amplitude from 0 to 5V.

TTL Wave: - Frequency from 1 Hz to 1 MHz in 5 Steps and Fixed Amplitude 5V.

- **Digital Input**

16 Digital Inputs Facility to Provide either Low or High Input (Slide Switch)

- **Two Single Stepping Inputs**

A and B with Inverting and Non Inverting Polarities

- **Adapter**

BNC to BS2 = 2

Banana CONNECTOR TO Bs2 = 2

- **Speaker**

One 8 Ohm Speaker

Two Seven Segment Display

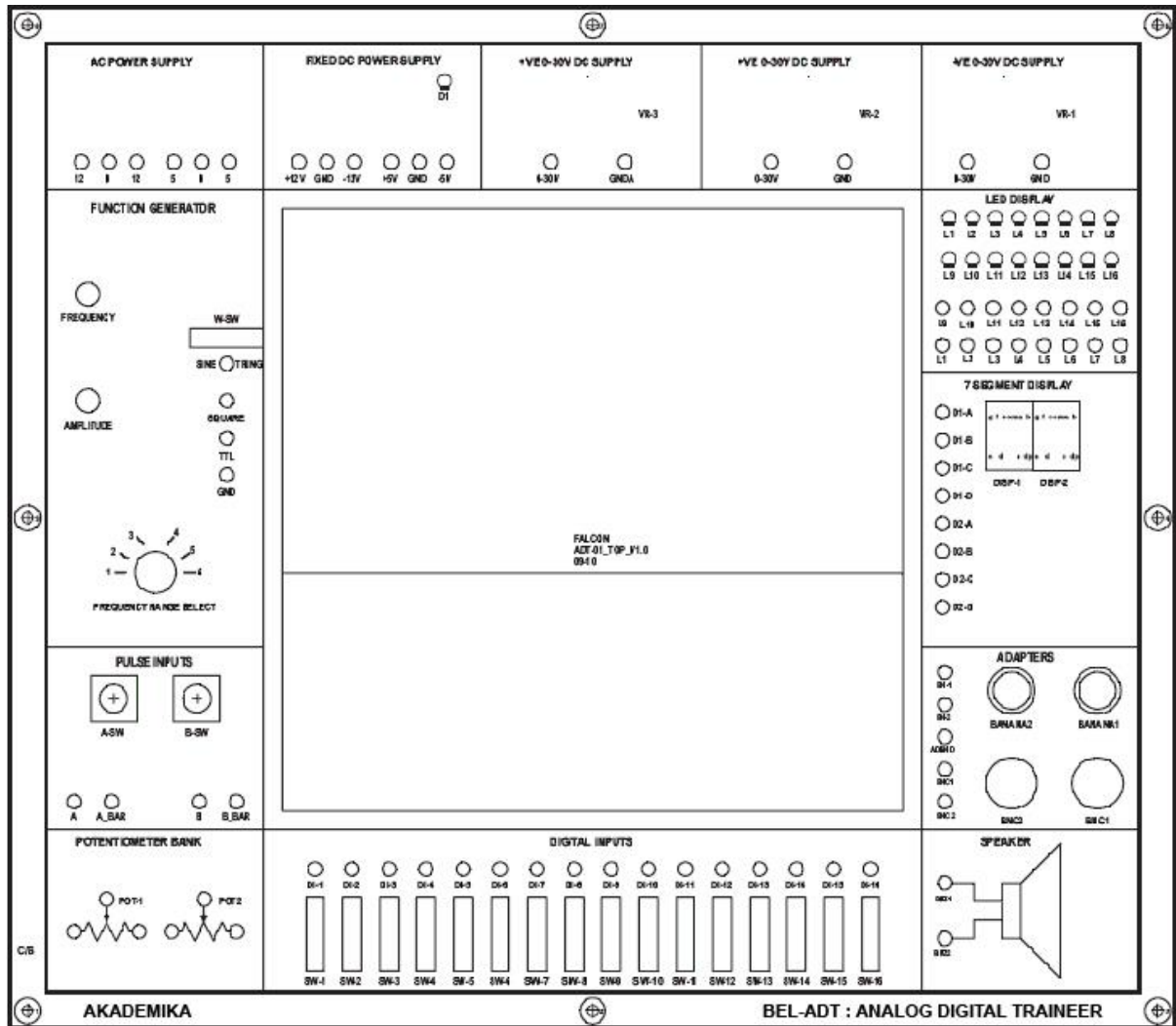
16 LED Display

One Bread Board (2 Power + 2 Sips), 4 ZIF Socket (40 Pin-1, 28 Pin-2, 20 Pin-1)

- **Accessories**

BEL-ADA			Quantity
1	Short Links PCS2 Patch cord	Male To Male	15
2	Long Links PCS2 Patch cord	Male To Male	5
3	EXPT Manual		1
4	Power Supply		1
5	Power cord		1

FUNCTIONAL BLOCK



INTRODUCTION

BEL-ADT is High Level, High Quality Analog-Digital Trainer kit. It combines all essential functions of Analog and Digital Experiment with removable breadboard, Includes DC power supply, Function Generator, Two Pulse Switches, 8 ohm 0.5W Speaker etc.

Analog signals equate levels of electric voltage or current to amounts of information by applying some rule. Consider for instance, an analog clock, where the passage of time is displayed as the motion of the clock hands. In electric signals, a certain amount of voltage corresponds directly to a measured physical phenomenon. For instance, on an accelerometer, the amount of acceleration, measured in g's will correspond directly to volts. So at one g, we have 1 volt output, at 2 g's we have 2 volts output, etc. Analog signals have an advantage that they can represent any fractional quantity, by outputting an equivalent fractional quantity of voltage or current. Put in a different manner an Analog signal is continuous in time and also continuous in value.

Digital simply means represented by digits. Think about how many times each day you encounter analog values that are represented with digits. The temperature is 79.8 degrees. The speed limit is 45 miles per hour, etc. Not surprisingly, digital electronics represents values with digits.

The term digital is also used when referring to binary devices such as the circuitry that makes a calculator work, the microprocessor in a computer, and the BASIC Stamp microcontroller. It's true - they are all digital devices. Binary devices are digital devices using two digits, 0 and 1.

HOW TO OPERATE THE BEL-ADT BOARD

1) Power Supply Section

BEL-ADT trainer kit provides on board power supply of following specifications.

DC power supply.

Fixed DC Output : +5V, +12V, -5V and -12V, 500mA

Variable DC Output : Dual DC 0 to 30V positive and single 0 to 30V negative, 500mA.

AC power supply : 5-0-5 and 12-0-12

A) AC Power Supply

Here 12-0-12 and 5-0-5 tabs of direct AC power supply are given. Step down transformer is used to convert AC230V to AC12V and 5V. User can directly uses the required AC power from these tabs using patch cord provided with the trainer kit.

B) Fixed DC Power Supply

Here +12V, GND, -12V and +5V, GND, -5V tabs of fixed DC power supply are given. Normal regulator IC's are used to convert AC supply into required fixed DC power supply. Power ON LED shows the +5V supply is working fine. User can directly uses the required fixed DC power from these tabs using patch cord provided with the trainer kit.

C) Variable DC Power Supply

Here VR1 is a negative power supply, VR2 is positive power supply and VR3 is an isolated GND positive power supply. All of these power supplies vary from 0 to 30V. User can set the required output voltage at the output post of respective power supply by varying respective power supply. I.e. to set the VR1 for 8V then vary the VR1 POT in clockwise direction or anticlockwise direction to get the output =8V at VR1 output post. User can directly uses the required DC power from these tabs using patch cord provided with the trainer kit.

Note: - All GND are same but VR3-GNDA is totally isolated from GND which is circuit gnd.

2) Onboard Function Generator

BEL-ADT trainer kit provides on board function generator of following specification.

Function generator

Wave forms : Sine, Triangular, square and TTL

Sine Wave : Frequency from 1 Hz to 1MHz in 5 steps and amplitude from 0 to 6Vp-p.

Triangular Wave: Frequency from 1 Hz to 800 KHz in 5 steps and amplitude from 0 to 6Vp-p.

Square Wave : Frequency from 1 Hz to 800 KHz in 5 steps and amplitude from 0 to 5V.

TTL Wave : Frequency from 1 Hz to 1 MHz in 5 steps and fixed amplitude 5V.

Function Generator has a direct output tab for TTL, square wave and GND output. There is one tab which is common for sine and triangular wave. Slide switch W-SW is used to either select Sine wave (keep left side) or Triangular wave (keep right side). User can set the frequency of wave form using “Frequency Range Select” switch and “Frequency” POT. Use “Frequency Range Select” switch in clock wise direction 5 steps to increase the frequency range. Use “Frequency” POT in anti-clock wise direction to increase the frequency. Use “Amplitude” POT in anti-clock wise direction to increase the amplitude of the selected frequency.

User can directly use the patch cord provided with the trainer kit for further connectivity.

3) Pulse Input Generator

BEL-ADT trainer kit provides on board pulse input generator of following specification.

Specifications

Push Buttons contain switches De-bouncer for eliminating the bounce caused by switch from “Open” to “Close” or from “Close” to “Open” position.

When user press the switch then output pulses are present on respective output tabs. Switch A is used to create positive going pulse at A tab and negative going pulse at A_bar tab. Switch B is used to create positive going pulse at B tab and negative going pulse at B_bar tab. User can directly use the patch cord provided with the trainer kit for further connectivity.

4) Digital Data Switch

BEL-ADT trainer kit provides on board 16 digital inputs of following specification.

Specifications:-

16 Toggle Switches and corresponding output point. When switch is set at “Down” position, the output is LOW level; contrarily, it is to be ‘High’ level while setting at “Up” position.

User can directly use the patch cord provided with the trainer kit for further connectivity.

5) Potentiometer Bank2

BEL-ADT trainer kit provides on board potentiometers of following specification.

Specifications:-

Potentiometers

Variable Resistor POT-01 : 1K Ω

Variable Resistor POT-02 : 10K Ω

User can directly use the patch cord provided with the trainer kit for further connectivity.

6) 16 Led Output's

BEL-ADT trainer kit provides on board 16 LED's for any analog light application or digital signal indication of following specification.

Specifications

BEL-ADT board has separate 16-LED's input terminal. The LED will be lighted up when input is at “High level”, and it will be turned OFF when it is at “No Input” or at “Low level”.

User can directly use the patch cord provided with the trainer kit for further connectivity.

7) Two Digit 7-Segment Display

BEL-ADT trainer kit provides on board two 7 segment display of following specification.
Specifications:-

BCD to 7 segment decoder is used with 7 segment display. Use the following truth table of BCD to 7 segment decoder IC to justify the functionality of display.

User can directly use the patch cord provided with the trainer kit for further connectivity.



DECIMAL OR FUNCTION	INPUTS						$\overline{BI}/\overline{RBO}^\dagger$	OUTPUTS						
	$\overline{LT}$	$\overline{RBI}$	D	C	B	A		a	b	c	d	e	f	g
0	H	H	L	L	L	L	H	ON	ON	ON	ON	ON	ON	OFF
1	H	X	L	L	L	H	H	OFF	ON	ON	OFF	OFF	OFF	OFF
2	H	X	L	L	H	L	H	ON	ON	OFF	ON	ON	OFF	ON
3	H	X	L	L	H	H	H	ON	ON	ON	ON	OFF	OFF	ON
4	H	X	L	H	L	L	H	OFF	ON	ON	OFF	OFF	ON	ON
5	H	X	L	H	L	H	H	ON	OFF	ON	ON	OFF	ON	ON
6	H	X	L	H	H	L	H	OFF	OFF	ON	ON	ON	ON	ON
7	H	X	L	H	H	H	H	ON	ON	ON	OFF	OFF	OFF	OFF
8	H	X	H	L	L	L	H	ON	ON	ON	ON	ON	ON	ON
9	H	X	H	L	L	H	H	ON	ON	ON	OFF	OFF	ON	ON
10	H	X	H	L	H	L	H	OFF	OFF	OFF	ON	ON	OFF	ON
11	H	X	H	L	H	H	H	OFF	OFF	ON	ON	OFF	OFF	ON
12	H	X	H	H	L	L	H	OFF	ON	OFF	OFF	OFF	ON	ON
13	H	X	H	H	L	H	H	ON	OFF	OFF	ON	OFF	ON	ON
14	H	X	H	H	H	L	H	OFF	OFF	OFF	ON	ON	ON	ON
15	H	X	H	H	H	H	H	OFF	OFF	OFF	OFF	OFF	OFF	OFF
BI	X	X	X	X	X	X	L	OFF	OFF	OFF	OFF	OFF	OFF	OFF
RBI	H	L	L	L	L	L	L	OFF	OFF	OFF	OFF	OFF	OFF	OFF
LT	L	X	X	X	X	X	H	ON	ON	ON	ON	ON	ON	ON

8) Channel Adaptor

BEL-ADT trainer kit provides on board two 7 segment display of following specification.
Specifications:-

Four Channel Adaptors

The two banana sockets' and two BNC jacks' point tips are changeable. It is suitable for the trainer to be connected with peripherals. Using this adapter facility user can connect outside world devices like function generator and oscilloscope for testing purpose to BEL-ADT trainer kit.

User can directly use the patch cord provided with the trainer kit for further connectivity.

9) Speaker

BEL-ADT trainer kit provides on board one speaker of following specification.

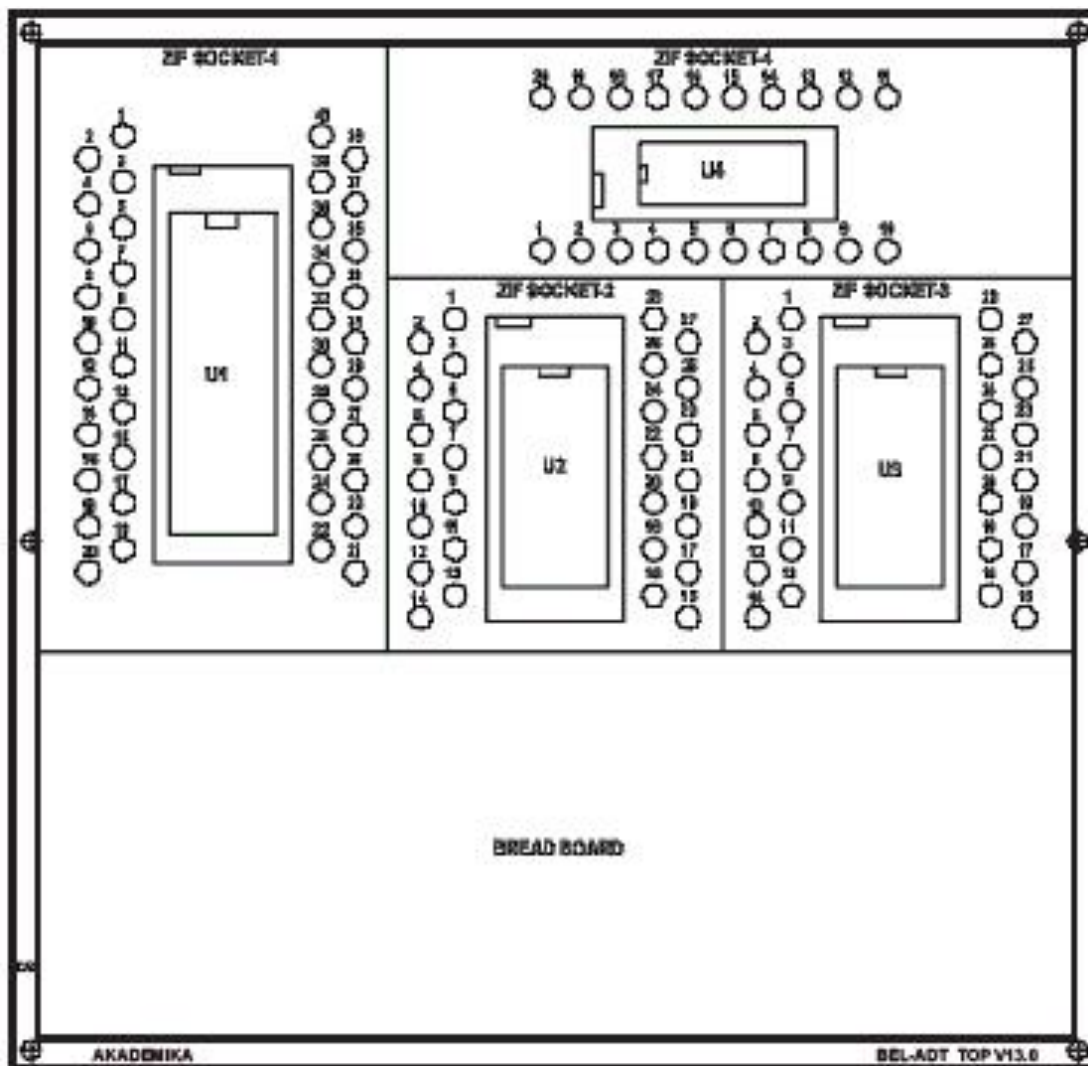
Specifications:-

One 8Ω/0.5W speaker

Used for loading purpose or for audio signal testing purpose. User can directly use the patch cord provided with the trainer kit for further connectivity.

BEL-ADT TOP BOARD

BEL-ADT Top Board provides 4-ZIF sockets and bread board to build basic level circuit and test it.



1) ZIF Sockets

4 ZIF Sockets (40 pin-1, 28 pin-2, 20 pin-1). User can insert any IC in the ZIF socket and test it using BEL-ADT board resources. User can directly use the patch cord provided with the trainer kit for further connectivity.

2) Single Bread Board

Interconnected with Tie points Nickel Plated Contact, fitted with all DIP sizes and all components with Lead and Solid Wire. It can be changed and replaced for different purpose and can be connected with demonstration panel. Therefore, it is very convenient for both teachers and students.

One bread board (2 power + 2 sips) User can build the basic level circuit and test it using BEL-ADT bread resources.